

Mass loss model for pebble aggregate samples in DiMES

Erick Martinez Loran

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1 The mass loss is proportional to the incident heat load

$$\frac{dm}{dt} = cq_{\parallel} \quad (1)$$

Consider the incident heat loss as

$$q_{\parallel} = \gamma n_e k_B T_e c_s, \quad (2)$$

where γ is the transmission coefficient in the sheath, k_B is Boltzmann's constant, c_s is the ion sound speed, T_e , and n_e are the electron temperature and electron density at the sheath edge, respectively.

The ion saturation current density J_{sat} is the maximum current that can be drawn to a material surface (such as a Langmuir probe or vacuum chamber wall) and is defined as follows:

$$J_{\text{sat}} \equiv en_{\text{se}}c_s \quad (3)$$

Substitute $n_{\text{se}}c_s = J_{\text{sat}}/e$ into the sheath transmission equation Eq. (2):

$$q_{\parallel} = \frac{\gamma J_{\text{sat}} k_B T_e}{e}. \quad (4)$$

Multiply Eq. (4) by the wetted area A_w and plug it into Eq. (1):

$$\frac{dm}{dt} = c A_w \frac{\gamma J_{\text{sat}} k_B T_e}{e} \quad (5)$$

We propose that $A_w(t)J_{\text{sat}}(t)$ is the current measured at DiMES:

$$\frac{dm}{dt} = c \gamma k_B I(t) T_e(t).$$

The prefactor $c \gamma k_B$ can be lumped into a single constant c

$$\boxed{\frac{dm}{dt} = c I(t) T_e(t)} \quad (6)$$

To calibrate the model, we measure the total mass change $m_L \equiv \Delta m = m_f - m_i$ before and after exposure to the plasma heat load. Then,

$$m_L = c \int_0^t I(t) T_e(t) dt = cQ. \quad (7)$$

So:

$$c = \frac{m_L}{Q} \quad (8)$$

2 Estimating the uncertainty in c

The total uncertainty is given by:

$$\delta c = \left[\left(\frac{\delta m_L}{m_L} \right)^2 + \left(\frac{\delta Q}{Q} \right)^2 \right]^{1/2} \quad (9)$$

2.1 Q is estimated numerically

Q is estimated using either the trapezoid or Simpson's rule.

Simpson's rule

$$Q = \frac{\Delta t}{3} \sum_{i=0}^N w_i I_i T_{e,i}, \quad w_i \in \{1, 4, 2, 4, 2, \dots, 4, 1\} \quad (10)$$

So each contribution is $q_i = \frac{\Delta t}{3} w_i I_i T_{e,i}$:

$$Q = \sum_{i=0}^N q_i$$

The partial derivatives of q_i are

$$\frac{\partial q_i}{\partial I_i} = \frac{q_i}{I_i}, \quad \frac{\partial q_i}{\partial T_{e,i}} = \frac{q_i}{T_{e,i}}$$

So

$$\begin{aligned} \delta q_i &= \left[\left(\frac{q_i \delta I_i^2}{I_i} \right)^2 + \left(\frac{q_i \delta T_{e,i}^2}{T_{e,i}} \right)^2 \right]^{1/2} \\ &= q_i \left[\left(\frac{\delta I_i^2}{I_i} \right)^2 + \left(\frac{\delta T_{e,i}^2}{T_{e,i}} \right)^2 \right]^{1/2} \end{aligned}$$

The uncertainty in Q is

$$\delta Q = \left[\sum_i \left(\frac{\partial Q}{\partial q_i} \right)^2 \delta q_i^2 \right]^{1/2} = \left[\sum_i \delta q_i^2 \right]^{1/2}$$

Then,

$$\delta Q = \left\{ \sum_i q_i^2 \left[\left(\frac{\delta I_i}{I_i} \right)^2 + \left(\frac{\delta T_{e,i}}{T_{e,i}} \right)^2 \right] \right\}^{1/2}$$

Expanding q_i :

$$\delta Q = \left\{ \sum_i \left(\frac{\Delta t}{3} w_i I_i T_{e,i} \right)^2 \left(\frac{\delta I_i}{I_i} \right)^2 + \sum_i \left(\frac{\Delta t}{3} w_i I_i T_{e,i} \right)^2 \left(\frac{\delta T_{e,i}}{T_{e,i}} \right)^2 \right\}^{1/2}$$

Considering the systematic uncertainty on the current: $\delta I_i / I_i \equiv \epsilon_I$ (from considering a $\epsilon_I = 5\%$ tolerance resistor and $I = V/R \Rightarrow \delta I = I \delta R / R = \epsilon_I I$), and canceling terms:

$$\delta Q = \left\{ \sum_i \left(\frac{\Delta t}{3} w_i T_{e,i} I_i \epsilon_I \right)^2 + \sum_i \left(\frac{\Delta t}{3} w_i I_i \delta T_{e,i} \right)^2 \right\}^{1/2}$$

The first term is equal to $\epsilon_I Q$:

$$\delta Q = \left\{ (\epsilon_I Q)^2 + \sum_i \left(\frac{\Delta t}{3} w_i I_i \delta T_{e,i} \right)^2 \right\}^{1/2} \quad (11)$$

3 Expanding the uncertainty in the rate of mass loss

The rate of mass loss is given by Eq. (6):

$$r(t) = \frac{dm}{dt} = c I(t) T_e(t). \quad (12)$$

The constant c , the current $I(t)$, and the electron temperature $T_e(t)$ have associated uncertainties δc , $\delta I = \epsilon_I$, and δT_e .

Writing Eq. (12) explicitly

$$r(t) = \frac{m_L}{\int_0^\tau I(t) T_e(t) dt} I(t) T_e(t).$$

Take the logarithm

$$\begin{aligned} \ln r(t) &= \ln(m_L) + \ln[I(t)] + \ln[T_e(t)] - \ln \left(\int_0^\tau I(t) T_e(t) dt \right) \\ &= \ln m_L + \ln I + \ln T_e - \ln Q \end{aligned} \quad (13)$$

Take the partial derivative

$$\frac{\partial \ln r}{\partial m_L} = \frac{1}{r} \frac{\partial r}{\partial m_L} = \frac{1}{m_L} \quad (14)$$

Varying each error source, **one at a time**:

3.1 Variation 1: δm_L (independent)

$$\left(\frac{\delta r}{r} \right)_{m_L} = \frac{1}{r} \left(\frac{\partial r}{\partial m_L} \right) \delta m_L = \frac{\delta m_L}{m_L}. \quad (8)$$

3.2 Variation 2: $\delta R/R$ (systematic, affects all I_i and $I(t)$)

Every I_i has propagates the uncertainty as $I_i \rightarrow I_i(1 + \epsilon)$ and $I(t) \rightarrow I(t)(1 + \epsilon)$.

We are assuming that the error is systematic: R has a **fractional error** $\epsilon = \delta R/R$.

Since $I = V/R$, a resistor that is **larger than assumed** means the **current is smaller than assumed**:

$$I_{\text{true}} = \frac{V}{R_{\text{true}}} = \frac{V}{R(1 + \epsilon)} \approx I(1 - \epsilon)$$

So strictly speaking $\delta I/I = -\epsilon$. However, for **uncertainty propagation, we only care about the magnitude** of the effect, not the sign. Writing $I \rightarrow I(1 + \epsilon)$, with $\epsilon = \pm \delta R/R$ is just a convenient way to track how a fractional shift propagates through the formula.

Then, $Q \rightarrow Q(1 + \epsilon)$:

$$Q = \int_0^\tau I(t)T_e(t)dt \rightarrow \int_0^\tau I(t)(1 + \epsilon)T_e(t)dt = (1 + \epsilon) \int_0^\tau I(t)T_e(t)dt = Q(1 + \epsilon).$$

This causes cancelation in r : Substitute both shifts in r :

$$r = cI(t)T_e(t) = \frac{m_L}{Q}I(t)T_e(t)$$

$$r \rightarrow \frac{m_L}{Q(1 + \epsilon)}I(t)(1 + \epsilon)T_e(t) = \frac{m_L}{Q}I(t)T_e(t) \frac{(1 + \epsilon)}{(1 + \epsilon)} = r$$

The $(1 + \epsilon)$ factors cancel **exactly**, so r is completely insensitive to ϵ

3.3 Variation 3: δT_e

Assuming Simpson's rule, $Q = \frac{\Delta t}{3} \sum_i w_i I_i T_{e,i}$. Then,

$$\ln r_k = \ln m_L + \ln I_k + \ln T_{e,k} - \ln \left(\frac{\Delta t}{3} \sum_j w_j I_j T_{e,j} \right) \quad (15)$$

T_e appears **twice**:

- Once **explicitly** in $+\ln T_{e,k}$
- Once **implicitly** inside $-\ln Q$, at every time step j , including $j = k$

Case $j \neq k$: $T_{e,j}$ appears only inside $\ln Q$

$$\frac{\partial \ln r_k}{\partial T_{e,j}} = -\frac{\partial \ln Q}{\partial T_{e,j}} = -\frac{1}{Q} \frac{\partial Q}{\partial T_{e,j}} = -\frac{w_j I_j \frac{\Delta t}{3}}{Q} \quad (9)$$

Case $j = k$: $T_{e,j}$ appears in both terms $\ln T_{e,k}$, and $-\ln Q$ simultaneously

$$\begin{aligned} \frac{\partial \ln r_k}{\partial T_{e,k}} &= \underbrace{\frac{\partial \ln T_{e,k}}{\partial T_{e,k}}}_{\text{explicit term}} - \underbrace{\frac{\partial \ln Q}{\partial T_{e,k}}}_{\text{implicit term}} \\ &= \frac{1}{T_{e,k}} - \frac{w_j I_j \frac{\Delta t}{3}}{Q} \end{aligned} \quad (16)$$

Multiplying through by $\delta T_{e,k}$ to express as a relative uncertainty:

$$\frac{\partial \ln r_k}{\partial T_{e,k}} \delta T_{e,k} = \left[1 - \frac{w_k I_k T_{e,k} \frac{\Delta t}{3}}{Q} \right] \frac{\delta T_{e,k}}{T_{e,k}} \quad (17)$$

Define

$$\alpha_k \equiv \frac{w_k I_k T_{e,k} \frac{\Delta t}{3}}{Q} \quad (18)$$

3.4 Full T_e uncertainty

We can unify both cases cleanly

$$\left(\frac{\delta r_k}{r_k} \right)_{T_e}^2 = \sum_j \left(\frac{\partial \ln r_k}{\partial T_{e,j}} \delta T_{e,j} \right)^2 = \sum_j \left(\frac{\partial}{\partial T_{e,j}} [\ln T_{e,k} - \ln Q] \delta T_{e,j} \right)^2 \quad (19)$$

where

$$\frac{\partial \ln r_k}{\partial T_{e,j}} = \begin{cases} \frac{1 - \alpha_k}{T_{e,k}}, & j = k \\ -\frac{w_j I_j \frac{\Delta t}{3}}{Q}, & j \neq k \end{cases} \quad (20)$$

So the sum over **all** j naturally splits into the two cases:

$$\left(\frac{\delta r_k}{r_k} \right)^2 = (1 - \alpha_k)^2 \left(\frac{\delta T_{e,k}}{T_{e,k}} \right)^2 + \sum_{j \neq k} \left(\frac{w_j I_j \frac{\Delta t}{3}}{Q} \right)^2 \delta T_{e,j}^2 \quad (21)$$